

Total No. of Questions : 8]

SEAT No. :

PC2374

[6354]-491

[Total No. of Pages : 2

B.E. (Computer Engineering)

OBJECT ORIENTED MODELING & DESIGN

(2019 Pattern) (Semester- VII) (Elective - III) (410244D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) What is an Actor, a Use Case, Use case diagram, Explain various relationships in use case diagram with example. [5]
- b) Draw State Diagram for cash Dispenser and explain concurrency. [5]
- c) Draw an ACTIVITY diagram for describing how selling two wheeler to a customer take place at a dealers Place. Customer can enquire about the two wheeler model available, a sale person is assign to show the customer the vehicles, the sale person show the available two wheelers, customer choose the model for purchase, he can optionally take, accessories like rear-guard etc,ask for a quotation and choose to purchase a vehicle if interested Explain notation used in a sentence each. [8]

OR

- Q2)** a) Write the Use case description for Buying a beverage [5]
- b) Model a SEQUENCE diagram for 'Rent a Car'. Use Case in an online web based car rental application. Here are some of the assumptions. The customer needs to first choose the type of the car he wants to rent. The car database is maintained in the system organized into type like family car, sports car etc. Based on the car available, the rates of rental are shown, the booking is then made, confirmed, the booking details stored in the system and user is issued an electronic confirmation of the booking. [5]
- c) Draw Sate diagram for telephone line connection. Give the meaning of Event, State and transition in the diagram. [8]

P.T.O.

- Q3)** a) Illustrate different reuse things that can be considered during modelling a system. [5]
 b) What are the different Software Control Strategy used in system design. [5]
 c) Illustrate what is Subsystem along with ways in which Subsystems can communicate. [7]

OR

- Q4)** a) While designing a system explain how data storage management is designed. [5]
 b) How Global Resources are Handled while System design [5]
 c) What are the different architecture styles followed in Software design. Draw Architecture of ATM system. [7]

- Q5)** a) How adjustment of inheritance is done in class design. Explain. [5]
 b) Explain the tasks involved in design optimization. [5]
 c) What is the importance of adjustment of inheritance? Discuss the steps doing it. [7]

OR

- Q6)** a) What are the steps involved in class design. Explain How to bridge the gap from high level requirement to low level services. [5]
 b) Discuss how you identify use cases and actors with respect to use case diagrams? [5]
 c) Explain the following terms in relation to class design [7]
 i) Refactoring
 ii) Reification.

- Q7)** a) Explain three important parts of design pattern. [5]
 b) What is communication pattern? Explain any one communication pattern in detail. [5]
 c) What is Design Pattern? Explain different types of design Patterns. [8]

OR

- Q8)** a) Explain counted pointer example in detailed. [5]
 b) What is view handler pattern? Explain in detail. [5]
 c) Write short note on [8]
 i) Client Dispatcher server
 ii) Publisher subscriber

